

Mohammad Haider

Software Engineer | Data systems, language models & simulation

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Software engineer with Microsoft and University of Virginia experience. Owns production data systems and migrations; builds language models, analytics products, and simulation software.

PROFESSIONAL EXPERIENCE

University of Virginia

May 2025 - Mar 2026

IT Analyst Intermediate, BI Developer

Charlottesville, VA

- Owned SQL-backed Tableau delivery from stakeholder interviews through deployment and maintenance; translated ambiguous operational questions into shared metric definitions.
- Reconciled purchase-order states across financial systems and built reporting that surfaced discrepancies and their financial impact.
- Reduced organization-wide dashboard load times by **approximately 20%** through collaborative, telemetry-driven optimization.
- Built environmental and economic-impact reporting for the ReUSE Store that reached **over 400 views**; adapted Tableau views as operational workflows changed.

Microsoft

Jul 2021 - Jul 2024

Software Engineer I

Redmond, WA

- Owned a **zero-downtime** SSAS Multidimensional-to-Tabular migration; defined parity checks, monitored **four weeks** of pipeline behavior, and kept the legacy service available until cutover criteria were met.
- Wrote and deployed Azure Data Factory verification scripts for daily pipeline monitoring and legacy-to-Tabular data consistency.
- Validated downstream reporting workflows showing **up to 200x** faster filtering and **approximately 4x** faster Excel loading.
- Built a full-stack C#/.NET prototype that guided configuration, generated configuration files, and validated output.

Microsoft

Summers 2017, 2018

Software Engineering Intern

Redmond, WA

- Built a C# service that identified each OneDrive user's most recently used items for a product UI feature; validated the service in development.

SELECTED ENGINEERING PROJECTS | INDEPENDENT

Indigo Circuit | Tournament analytics platform

indigocircuit.app

- Launched in April**; sustained automatic updates **within 24 hours** of every major tournament on database-hosting costs alone, then redesigned the frontend with progressive disclosure.
- Reduced refresh-related read blocking from ~9 minutes to under 1 second through shadow builds and atomic publication; preserved live data through two observed production rebuild failures.

LM Lab | Language-model engineering

github.com/haidmoham/lmlab

- Implemented an autoregressive PyTorch stack with BPE, multi-head causal attention, residual blocks, and learned/sinusoidal positions; completed **39 training runs** across three controlled studies.
- Measured 1- to 8-block models at 10,000 updates: mean held-out cross-entropy fell from **3.529 to 3.337 nats/token**; verified checkpoint restoration and recorded compute costs.

C-1N + Robotics Test Bench | Simulation & control software

mhaider.dev/c1n

- Expanded a MuJoCo robot from **12 to 18 actuated joints**; unified simulation, replay, and telemetry; built a deterministic 10-second standing baseline and a **33-case disturbance suite**.

TECHNICAL SKILLS

Languages: Python, SQL, C#, R

Production: C#/.NET, Azure Data Factory, SSAS Tabular, Azure Synapse, Kusto, Tableau

Project stack: PyTorch, DuckDB, dbt, Flask, MuJoCo, NumPy, pandas, Jupyter, Git, GitHub Actions

EDUCATION

Virginia Tech | B.S. in Statistics

2021